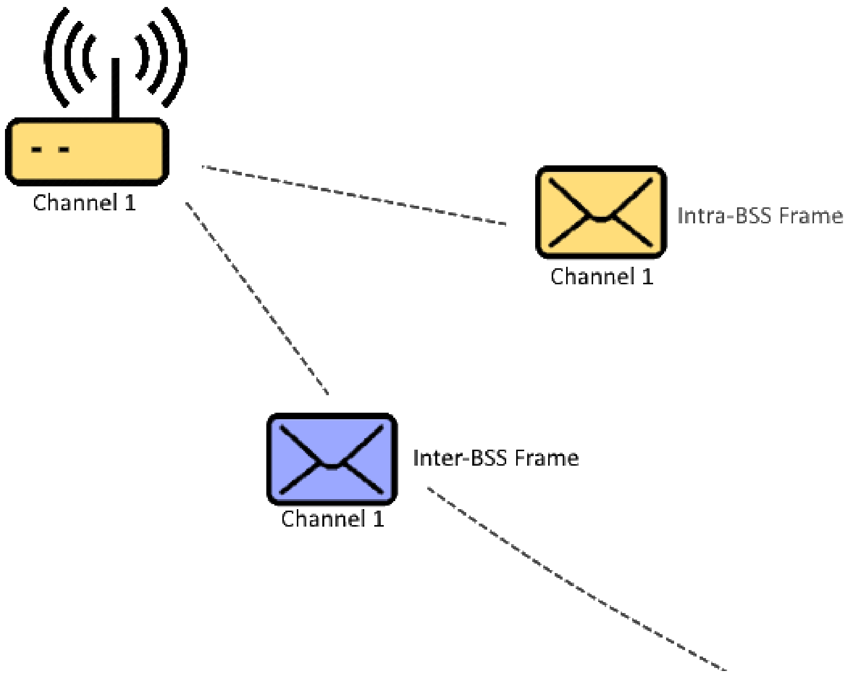
BSS Coloring in Wi-Fi 6

Improving Spatial Reuse in Dense Wireless Networks



Why Do We Need BSS Coloring?

Multiple Wi-Fi networks may:

- Operate on the same channel
- Have overlapping coverage
- Detect each other's transmissions
- Unnecessarily defer transmission

The Traditional Wi-Fi Problem

Listen Before Transmit

CSMA/CA + CCA

Want to transmit

↓

Check channel

↓

Is it busy?

↙ ↘

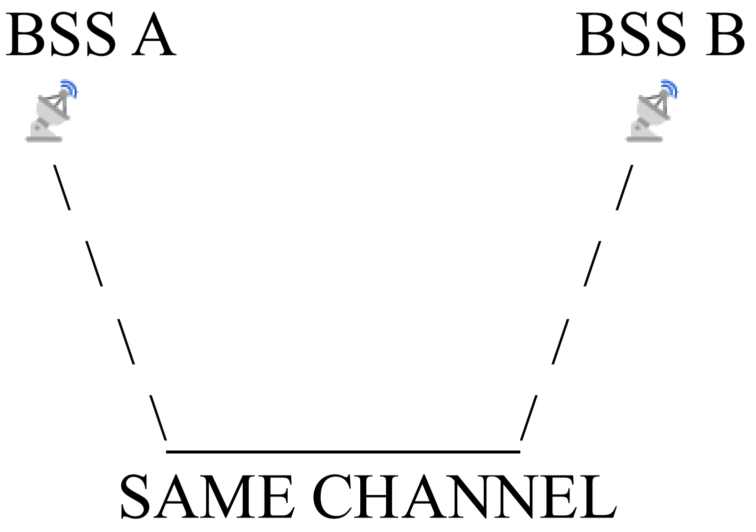
YES NO

↓ ↓

Wait Transmit

The Spatial Reuse Problem

What if the neighbouring signal is weak?



BSS A is transmitting
BSS B detects it
But: Is the signal actually strong enough to cause harmful interference?

What Is Spatial Reuse?

Reuse the same channel in different locations when interference conditions allow it.

AP-A ——————————→ Client A

AP-B ——————————→ Client B

Same channel + different spatial locations

Introducing BSS Coloring

Each 802.11ax BSS is associated with a: 6-bit BSS Color

Usable values: 1–63

BSS A → Color 12

BSS B → Color 34

Where Is the BSS Color?

Inside an 802.11ax HE Transmission

```
┌──────────────────────────────┐
│ HE PHY Signaling             │
│                              │
│ BSS Color                    │
│ Other HE parameters          │
├──────────────────────────────┤
│ MAC Header                   │
├──────────────────────────────┤
│ Data                         │
└──────────────────────────────┘
```

https://wifisharks.com/2023/06/04/11ax-bss-color/

Own BSS vs OBSS

BSS Color helps distinguish transmissions

My BSS Color = 15

Received Color = 15

↓

MY BSS

My BSS Color = 15

Received Color = 37

↓

OBSS

BSS Coloring + OBSS-PD (Overlapping BSS Packet Detection)

How Spatial Reuse Becomes Possible

Signal detected
↓
Read BSS Color
↓
Own BSS or OBSS?
↓
OBSS
↓
OBSS-PD
↓
Check signal conditions
↓

Weak	Strong
↓	↓
Potential reuse	Defer

For an OBSS transmission:

Received OBSS signal
↓
Compare against
OBSS-PD conditions
↓

Weak enough	Too strong
↓	↓
Potential reuse	Defer

BSS Color Collision

BSS Colors Are Not Globally Unique
Example:
AP-A → Color 20
AP-B → Color 20

If neighboring BSSs use the same color: BSS Color Collision
802.11ax provides mechanisms to detect and resolve color conflicts.

Best suited for dense Wi-Fi environments

- 🏢 Enterprise offices
- 🎓 Universities
- 🏟️ Stadiums
- ✈️ Airports
- 🏠 Apartment buildings

Benefits

- Better spatial reuse
- Better channel utilization
- Higher aggregate capacity
- Useful in dense deployments
- Helps distinguish own BSS and OBSS

Limitations / Considerations

- Does not eliminate interference
- Requires 802.11ax support for the relevant features
- Color collisions can occur
- Threshold selection matters
- RF planning is still importa

Thankyou !!